

Miguel González

miguelgg.com
linkedin.com/in/miguelgonzalezgonzalez

mgonzalez.contacto@gmail.com
miguel.gonzalez@icmat.es

Research interests

Complex and algebraic geometry, moduli spaces of geometric structures on Riemann surfaces, Higgs bundles, mirror symmetry, Langlands duality.

Experience

PhD Research - <i>la Caixa</i> foundation InPhINIT Retaining Fellowship <i>Instituto de Ciencias Matemáticas (ICMAT)</i> <i>Advisers: Oscar García-Prada and Tamás Hausel</i>	2023- <i>Project: Classification of very stable Higgs bundles</i> <i>Grant LCF/BQ/DR23/12000030</i>
Research Grant <i>Instituto de Ciencias Matemáticas (ICMAT)</i> <i>Supervised by Oscar García-Prada</i>	Sept. 2022 - Jun. 2023 <i>Project: Cyclic Higgs bundles and Vinberg theory</i> <i>Grant JAEICU-21-ICMAT-01, Top 1 application score</i>
Research Intern <i>Hausel Group (Geometry and its Interfaces), Institute Of Science And Technology Austria (ISTA)</i> <i>Supervised by Tamás Hausel</i>	Jul. 2022 - Aug. 2022 <i>Project: Very stable $U(p, q)$-Higgs bundles</i>
Collaboration grant <i>Department of Mathematics, Universidad Autónoma de Madrid</i> <i>Supervised by Ernesto Girondo Sirvent</i>	Nov. 2021 - Jul. 2022 <i>Project: Theoretical and computational research about properties of fundamental domains of Fuchsian groups</i>
Introduction to Mathematical Research Grant Severo Ochoa <i>Instituto de Ciencias Matemáticas (ICMAT)</i> <i>Supervised by Oscar García-Prada</i>	Summer 2021 <i>Project: Introduction to Moduli Spaces and Higgs Bundles</i>

Education

PhD Student in Mathematics Research <i>Universidad Complutense de Madrid (UCM) - Instituto de Ciencias Matemáticas (ICMAT)</i> <i>Advisers: Oscar García-Prada and Tamás Hausel</i>	2023-
Masters Degree in Mathematics <i>Universidad Autónoma de Madrid (UAM)</i> <i>Dissertation: Cyclic Higgs bundles and Vinberg theory</i>	2022 – 2023 <i>Average grade: 9.94 / 10</i> <i>5 courses with honours</i>
Double Degree in Mathematics and Computer Science <i>Universidad Autónoma de Madrid (UAM)</i>	2017 – 2022 <i>Average grade: 9.82 / 10</i> <i>44 courses with honours</i>

Preprints

Very stable parabolic G -Higgs bundles and affine flag varieties	June 2026 <i>arXiv:2606.16880</i>
Very stable regular G -Higgs bundles	March 2025 <i>arXiv:2503.01289</i>

Publications

Cyclic Higgs bundles and the Toledo invariant <i>With Oscar García-Prada</i>	March 2026 <i>Transformation Groups</i> , Vol. 31, 1505–1542 doi:10.1007/s00031-026-09962-2
Hitchin map on even very stable upward flows <i>With Tamás Hausel</i>	April 2024 <i>International Journal of Mathematics</i> , Vol. 35, No. 9 doi:10.1142/S0129167X2441009X

Talks

Very stable parabolic G -Higgs bundles and affine flag varieties <i>Workshop: Unifying Geometric Perspectives on Hitchin's Moduli Space</i> <i>Bernoulli Center, École polytechnique fédérale de Lausanne (EPFL)</i>	August 2026
Wobbly Higgs bundles and the Toledo invariant <i>i-Link geometry workshop</i> <i>Institute of Science and Technology Austria (ISTA)</i>	April 2026
Wobbly parabolic G -Higgs bundles and affine flag varieties <i>Geometric Structures and Stability Thematic Program</i> <i>International Centre for Theoretical Sciences (ICTS-TIFR)</i>	February 2026
Very stable G -Higgs bundles and Hecke modifications <i>MIST Conference in Mathematical Physics</i> <i>The Chinese University of Hong Kong (CUHK)</i>	November 2025
The structure of symmetry: an introduction to root systems <i>ICMAT-UAM Junior Colloquium</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	September 2025
A criterion for wobbliness of parabolic G -Higgs bundles <i>Seminar talk</i> <i>Chern Institute of Mathematics, Nankai University (CIM)</i>	August 2025
Compact components in relative character varieties for real lie groups of Hermitian type <i>Geometry Workshop of the Hitchin–Ngô Laboratory</i> <i>Feishu (Shanghai)</i>	August 2025
$\mathbb{C}^\times$ -actions, lagrangian submanifolds and very stable bundles <i>Workshop: Geometry and Topology for the Future III</i> <i>Universidad Complutense de Madrid (UCM)</i>	June 2025
Very stable regular G -Higgs bundles <i>Hitchin–Ngô Laboratory Special Activity</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	June 2025
Very stable regular G -Higgs bundles <i>Séminaire d'Analyse et Géométrie</i> <i>Institut de Mathématiques de Jussieu–Paris Rive Gauche (IMJ-PRG)</i>	May 2025
Very stable regular G -Higgs bundles <i>Conference: Stratifications of Higgs bundle moduli spaces and related topics</i> <i>Universidade de Santiago de Compostela (USC)</i>	May 2025
Hitchin map on minuscule Lagrangians <i>Bonn–Vienna block seminar: Equivariant cohomology, stable envelopes and big algebras</i> <i>University of Bonn</i>	February 2025
The decomposition theorem I: finite fields <i>Reading seminar on perverse sheaves</i> <i>Universidad Complutense de Madrid (UCM)</i>	January 2025
Very stable G -Higgs bundles with regular nilpotent Higgs field <i>RTGF (Spanish Geometry and Physics Thematic Network) meeting</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	November 2024
The $\mathbb{C}^\times$ -action on the moduli space of Higgs bundles <i>“Doc in progress” seminar</i> <i>Università degli Studi di Trento (UniTrento)</i>	November 2024
The classification of extended 2-TQFTs <i>Reading seminar on ∞-categories</i> <i>Universidad Complutense de Madrid (UCM)</i>	May 2024
Mixed Hodge theory <i>Arbeitsgemeinschaft: Geometry and Representation Theory around the $P=W$ Conjecture</i> <i>Mathematisches Forschungsinstitut Oberwolfach (MFO)</i>	April 2024

Cyclic Higgs bundles and the Toledo invariant <i>Hausel group seminar</i> <i>Institute of Science and Technology Austria (ISTA)</i>	January 2024
Introduction to Hodge theory in complex algebraic geometry <i>PhD seminar</i> <i>Universidad complutense de Madrid (UCM)</i>	January 2024
The Hitchin Fibration and Spectral Curves <i>Hitchin-Ngô Lab Working Group</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	February 2023

Attended conferences

Workshop: Unifying Geometric Perspectives on Hitchin's Moduli Space <i>Bernoulli Center, École polytechnique fédérale de Lausanne (EPFL)</i>	10-14 August 2026
2nd MaPhyAG Workshop: Algebraic Geometry and Mathematical Physics <i>Universidad Complutense de Madrid (UCM)</i>	29 June - 3 July 2026
i-Link geometry workshop <i>Institute of Science and Technology Austria (ISTA)</i>	7-10 April 2026
Geometric Structures and Stability Thematic Program <i>International Centre for Theoretical Sciences (ICTS-TIFR)</i>	16-27 February 2026
Hong Kong Laureate Forum 2025 <i>Hong Kong Science Park</i>	5-8 November 2025
MIST Conference in Mathematical Physics <i>The Chinese University of Hong Kong (CUHK)</i>	31 October-4 November 2025
Meeting of the Spanish Network of Geometry and Physics 2025 <i>Universitat de Barcelona (UB)</i>	15-17 September 2025
Geometry Workshop of the Hitchin–Ngô Laboratory <i>Feishu (Shanghai)</i>	7-9 August 2025
Feishu China Mathematics Symposium <i>Feishu (Shanghai)</i>	4-6 August 2025
Higher rank geometric structures, Higgs bundles and physics <i>Institut Henri Poincaré (IHP)</i>	19-23 May 2025
Stratifications of Higgs bundle moduli spaces and related topics <i>Universidade de Santiago de Compostela (USC)</i>	5-9 May 2025
Geometry at Large, an ICMAT–IMSA collaboration <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	21-25 April 2025
RTGF (Spanish Geometry and Physics Thematic Network) meeting <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	26-29 November 2024
1st MaPhyAG Workshop: Integrability, Geometry and QFT <i>Universidad Complutense de Madrid (UCM)</i>	7-11 October 2024
Clay Research Workshop: The P=W Conjecture in Non Abelian Hodge Theory <i>Mathematical Institute (University of Oxford)</i>	30 September-4 October 2024
2nd Simons Math Summer Workshop: Moduli <i>Simons Center for Geometry and Physics (SCGP)</i>	1-19 July 2024
Arbeitsgemeinschaft: Geometry and Representation Theory around the P=W Conjecture <i>Mathematisches Forschungsinstitut Oberwolfach (MFO)</i>	1-5 April 2024
The Hitchin system, Langlands duality and mirror symmetry <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	24-28 April 2023
Moduli spaces and geometric structures <i>Conference in honour of Oscar García-Prada on the occasion of his 60th birthday</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	12-16 September 2022

Poster presentations

Very stable G -Higgs bundles <i>Hong Kong Laureate Forum 2025</i> <i>Hong Kong Science Park</i>	November 2025
Very stable $U(p, q)$ -Higgs bundles <i>Summer ISTerns poster session</i> <i>Institute of Science and Technology Austria (ISTA)</i>	August 2022

Teaching

Linear algebra (Álgebra lineal) <i>Linear algebra problem sessions for first year undergraduate students in mathematics.</i> <i>Universidad Complutense de Madrid (UCM)</i>	Year 2025-2026, 1st semester
Pequeño Instituto de Matemáticas (PIM) / Little mathematics institute <i>Advanced mathematics problem-solving classes for high school students. https://www.icmat.es/PIM/</i> <i>Instituto de Ciencias Matemáticas (ICMAT)</i>	2023-2026

Research visits

Institute of Science and Technology Austria (ISTA) <i>Hausel Group (Geometry and its Interfaces)</i>	15 March - 17 May 2026
Institut Henri Poincaré, Paris <i>Higher rank geometric structures research program</i>	12-23 May 2025
Faculty of Mathematics, University of Duisburg-Essen <i>Heinloth Group</i>	7-11 April 2025
Institute of Science and Technology Austria (ISTA) <i>Hausel Group (Geometry and its Interfaces)</i>	23 February - 22 March 2025
Institute of Science and Technology Austria (ISTA) <i>Hausel Group (Geometry and its Interfaces)</i>	15-26 January 2024
Institute of Science and Technology Austria (ISTA) <i>Hausel Group (Geometry and its Interfaces)</i>	19-28 January 2023

Awards and prizes

Master's degree Extraordinary Award (best academic record) <i>Awarded by Universidad Autónoma de Madrid (UAM)</i>	2024
UAM Young Talent Award <i>Awarded by Universidad Autónoma de Madrid (UAM)</i>	2022
Best undergraduate academic record <i>Awarded by Universidad Autónoma de Madrid (UAM)</i>	2022
Excellence Grant <i>Awarded by the Comunidad de Madrid</i>	2018, 2019, 2020, 2021
Best academic record of first year undergraduate students <i>Awarded by Universidad Autónoma de Madrid (UAM)</i>	2018
Concurso de Primavera de Matemáticas (Mathematics contest) <i>Winner (Top 20)</i>	2017
Extraordinary Award of Bachillerato <i>Awarded by the Comunidad de Madrid to Top <1% high secondary school students</i>	2017
Extraordinary Award of Secondary Education <i>Awarded by the Comunidad de Madrid to Top <1% basic secondary school students</i>	2015

Miscellaneous publications

Report for Arbeitsgemeinschaft: Geometry and Representation Theory around the $P=W$ Conjecture

Editor and contributor with the extended abstract for my talk on Mixed Hodge theory.

doi:10.14760/OWR-2024-16

Undergraduate course notes

LaTeX notes for every course I took during my undergraduate mathematics studies.

Available at miguelgg.com

Declined grants (incompatibilities)

PhD studies grant

2023

FPU Grant - Spanish Ministry of Universities

Funding for a 4 year PhD research and teaching contract

PhD studies grant

2023

ICMAT Severo Ochoa FPI Grant - Spanish Ministry of Science and Innovation

Funding for a 4 year PhD research contract

Masters studies grant

2022-2023

Department of Mathematics, Universidad Autónoma de Madrid (UAM)

10000€ for masters studies and research

Top 1 application score

Skills

Spanish: Native level

English: Fluent level

French: Intermediate level

Japanese: Intermediate level (comprehension), basic level (production)

$\text{\LaTeX}$ knowledge

Computer skills: Python, C, Java, GitHub, Databases (SQL), Machine Learning and AI (Master rank in kaggle.com), among others.